

Abstract No.

Ventricular proximity defines a distinct spatial multi-omic landscape in glioblastoma

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Glioblastoma is characterised by marked molecular and microenvironmental heterogeneity, and tumour location may contribute to this variability. Tumours located close to the ventricular system have been associated with poorer clinical outcome, although the biological mechanisms underlying this relationship remain unclear. We investigated whether ventricular proximity is associated with distinct clinical, transcriptomic, genomic and proteomic features. A retrospective cohort of 57 patients with glioblastoma was stratified according to tumour distance from the ventricular system into close, intermediate and far groups. Overall survival was assessed using Kaplan–Meier analysis. Multi-omic characterisation included RNA sequencing and whole-genome sequencing in 23 tumours and quantitative LC–MS/MS proteomics in nine tumours. Differential expression, pathway enrichment, immune and tumour-state deconvolution, somatic variant analysis and functional protein annotation were used to compare spatial groups. Tumour proximity to the ventricular system was significantly associated with poorer overall survival. Close tumours displayed a stronger immune and inflammatory transcriptomic profile, with enrichment of antigen presentation, MHC class II signalling, leukocyte activation, cytokine-related processes and extracellular matrix remodelling. They also showed increased expression of immune checkpoint and T-cell dysfunction markers, together with hypoxia, epithelial-to-mesenchymal transition and mesenchymal-like programmes. Deconvolution analyses supported spatial differences in immune composition and glioblastoma cellular states. Whole-genome analysis identified promoter-region variants affecting immune and regulatory genes. Proteomic profiling further showed enrichment of complement, coagulation, antigen presentation and phagocytosis-related processes in close tumours, whereas distant tumours displayed greater representation of neuronal and synaptic functions. These findings indicate that ventricular proximity is associated with a distinct multi-omic phenotype characterised by immune activation and dysfunction, hypoxia, mesenchymal features and extracellular matrix remodelling. Anatomical location therefore appears to contribute to glioblastoma heterogeneity and may help explain the poorer prognosis observed in periventricular tumours.



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